

YGGDRASIL

A Constitutional DNS Layer for User Safety, Privacy, and the Sovereign Internet

Jeremy Karrick — Project-AI — 2026

Abstract

The Domain Name System (DNS) is foundational yet structurally vulnerable. This work introduces Yggdrasil as a compositional architecture enforcing verification, privacy, and adaptive defense at the kernel boundary.

This document integrates original text with truth-aligned system clarification without altering author intent.

Author's Note on System Positioning

Yggdrasil is a compositional architecture, not a completed system. Components vary in maturity and are clarified in Appendix A.

Part One: The Wound in the Phonebook

1.1 How DNS Works

DNS operates through recursive queries across resolvers to authoritative servers, ultimately anchored by root servers.

The protocol lacks authentication, encryption, and memory.

1.2 The Attack Surface

Cache poisoning, BGP hijacking, and passive observation are active and documented threats.

The Completeness of the Problem

DNS lacks verification, privacy, and memory. These are not separate failures but manifestations of the same design gap.

Scope Clarification

Yggdrasil does not replace DNSSEC, DoH, or DoT. It repositions enforcement.

Part Two: Architecture

System Model

$$Y = (I, V, S, M, F)$$

I: Interception, V: Verification, S: Shadow, M: Memory, F: Federation

Kernel Interception

eBPF-based interception at TC/XDP enables early enforcement but is constrained by verifier limits and kernel risk.

```
int dns_intercept(packet) {
    if (is_dns(packet)) {
        if (is_suspicious(packet)) {
            redirect_to_shadow(packet);
            return DROP;
        }
    }
    return PASS;
}
```

Verification

Multi-layer validation includes DNSSEC, source validation, and anomaly scoring.

$$\text{score} = \sum(\text{weight}_i * \text{feature}_i)$$

```
if score < T1: VALID
elif score < T2: SUSPICIOUS
else: INVALID
```

Shadow Plane

Suspicious traffic is redirected to controlled environments for observation instead of denial.

Memory System

Events are logged, weighted, and decayed over time.

```
Event {
    source_ip
    domain
    anomaly_type
    timestamp
    weight
}
```

Federation

Nodes share threat intelligence via gossip. This introduces poisoning risk without consensus mechanisms.

Part Three: Implementation Path

Components exist at prototype level with gaps in DNSSEC completeness, federation security, and real-world validation.

Status Summary

DNS: REAL Interception: PROTOTYPE Verification: PROPOSED Shadow: PROPOSED Memory: PROTOTYPE Federation: SPECULATIVE

Conclusion

Yggdrasil consolidates DNS defenses at the kernel boundary. It is compositional, not final.

Appendix A — Truth Alignment

REAL: validated PROTOTYPE: implemented but incomplete PROPOSED: defined SPECULATIVE: research